| | |
|---|---|
| CUSTOMER: 327510 | ORDER #: 2058213 |
| BATCH #: 1750593 | DELIVERY ID: 13081464 |
| PICK ZONE: LTO2 | PICK SEQUENCE #: 15500 |
| PRODUCT NAME: MB-215/ANCOCIDE 4070 | BARCODE #: 10189589 |

# Material Safety Data Sheet: MB-215/ANCOCIDE 4070

**Supercedes Date** 12/13/2011 **Issuing Date** 07/09/2013

## 1. PRODUCT AND COMPANY IDENTIFICATION

**Product Name** MB-215/ANCOCIDE 4070
**Recommended use** Biocidal product
**Information on Manufacturer**
CHEMSEARCH DIV. OF NCH CORP.
BOX 152170
IRVING, TX 75015

**Product Code** 10189589
**Chemical nature** Aqueous solution
**Emergency Telephone Number**
CHEMTREC® 800-424-9300

## 2. HAZARDS IDENTIFICATION

**Emergency Overview**
DANGER
Corrosive
Causes skin and eye burns
May cause allergic skin reaction
May cause delayed lung injury and burns
Harmful or fatal if swallowed

**Color** Colorless - Yellow **Physical State** Liquid **Odor** Pungent

| | |
|---|---|
| **Potential Health Effects** | |
| **Principle Route of Exposure** | Skin contact, Eye contact, Inhalation. |
| **Primary Routes of Entry** | Skin Absorption |
| **Acute Effects** | |
| **Eyes** | Corrosive to the eyes and may cause severe damage including blindness. |
| **Skin** | Causes skin burns. May cause allergic skin reaction. |
| **Inhalation** | Harmful by inhalation. Causes burns. Risk of serious damage to the lungs (by inhalation). |
| **Ingestion** | If ingested, severe burns of the mouth and throat, as well as a danger of perforation of the esophagus and the stomach. Components of the product create formation of methemoglobin. |
| **Chronic Toxicity** | May cause sensitization by skin contact. Inhaled corrosive substances can lead to a toxic edema of the lungs. |
| **Target Organ Effects** | Immune system, Blood. |
| **Aggravated Medical Conditions** | Skin disorders, Respiratory system. |
| **Potential Environmental Effects** | See Section 12 for additional Ecological information. |

## 3. COMPOSITION / INFORMATION ON INGREDIENTS

| Component | CAS-No |
|---|---|
| Magnesium nitrate | 10377-60-3 |
| 5-Chloro-2-methyl-4-isothiazolin-3-one | 26172-55-4 |
| Magnesium chloride | 7786-30-3 |

## 4. FIRST AID MEASURES

| | |
|---|---|
| **General advice** | Do not get in eyes, on skin or on clothing. Do not breathe mist. |
| **Eye Contact** | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses. Call a physician or poison control center immediately. |
| **Skin Contact** | Remove immediately all contaminated clothing. Wash off immediately with plenty of water for at least 15 minutes. Contact a poison control center. |
| **Inhalation** | Move to fresh air. If not breathing, give artificial respiration. Get medical attention immediately. |
| **Ingestion** | Call a physician or poison control center immediately. Give small amounts of water to drink. DO NOT induce vomiting unless directed to do so by a physician or poison control center. |
| **Notes to physician** | The product causes burns of eyes, skin and mucous membranes. Control of circulatory system, shock therapy if needed. Since reversion of methemoglobin to hemoglobin occurs spontaneously after termination of exposure, moderate degrees of cyanosis need to be treated only by supportive measures. May cause sensitization of susceptible persons. |

## 5. FIRE-FIGHTING MEASURES

**Flash Point** Does not flash **Method** Not applicable
**Autoignition Temperature** No information available.
**Flammability Limits in Air %** Hydrogen, by reaction with metals. **Upper** 75 **Lower** 4
**Suitable Extinguishing Media**
Foam. Alcohol-resistant foam. Carbon dioxide ($CO_2$). Dry chemical. Water spray. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
**Specific hazards arising from the chemical**
Thermal decomposition can lead to release of irritating gases and vapors. Contact with metals liberates flammable hydrogen gas. Material can create slippery conditions.
**Protective Equipment and Precautions for Firefighters**
As in any fire, wear self-contained breathing apparatus pressure -demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

| | | | |
|---|---|---|---|
| **NFPA** | **Health** 3 | **Flammability** 1 | **Instability** 0 |
| **HMIS** | **Health** 3 | **Flammability** 1 | **Instability** 0 |

## 6. ACCIDENTAL RELEASE MEASURES

**Personal Precautions** Use personal protective equipment. Ensure adequate ventilation. Prevent further leakage or spillage if safe to do so. Material can create slippery conditions.
**Environmental Precautions** Do not flush into surface water or sanitary sewer system.
**Methods for Containment** Contain spillage, soak up with non -combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and transfer to a container for disposal according to local / national regulations (see section 13)
**Methods for Cleaning Up** Pick up and transfer to properly labeled containers.
**Neutralizing Agent** Neutralize with lime milk or soda and flush with plenty of water.

## 7. HANDLING AND STORAGE

**Handling** Ensure adequate ventilation. Wear personal protective equipment.
**Storage** Keep out of the reach of children. Store in original container. Keep containers tightly closed in a cool, well-ventilated place. Metal containers must be lined. Freezing will affect the physical condition but will not damage the material. Thaw and mix before using.
**Storage Temperature** **Minimum** 34 °F / 1 °C **Maximum** 131 °F / 55 °C
**Storage Conditions** **Indoor** X **Outdoor** **Heated** **Refrigerated**

## 8. EXPOSURE CONTROLS / PERSONAL PROTECTION

**Exposure Guidelines**

| Component | ACGIH TLV | OSHA PEL | NIOSH |
|---|---|---|---|
| Magnesium nitrate | No data available | No data available | No data available |
| 5-Chloro-2-methyl-4-isothiazolin-3-one | No data available | No data available | No data available |
| Magnesium chloride | No data available | No data available | No data available |

**Engineering Measures** Ensure adequate ventilation, especially in confined areas. Where reasonably practicable this should be achieved by the use of local exhaust ventilation and good general extraction.
**Personal Protective Equipment**
**Eye/Face Protection** Tightly fitting safety goggles. Face-shield.
**Skin Protection** Wear suitable protective clothing, Impervious gloves.
**Respiratory Protection** In case of inadequate ventilation wear respiratory protection. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
**General Hygiene Considerations** Wear protective gloves/clothing. Ensure that eyewash stations and safety showers are close to the workstation location. Remove and wash contaminated clothing before re -use.

## 9. PHYSICAL AND CHEMICAL PROPERTIES

| | | | |
|---|---|---|---|
| **Physical State** | Liquid | **Viscosity** | Non viscous |
| **Color** | Colorless - Yellow | **Odor** | Pungent |
| **Appearance** | Transparent | **pH** | 2 |
| **Specific Gravity** | 1.02 | **Evaporation Rate** | 0.58 (Butyl acetate=1) |
| **Percent Volatile (Volume)** | 97.9 | **VOC Content (%)** | 0 |
| **VOC Photoreactive (Y/N)** | No | **VOC Content (g/L)** | 0 |
| **Vapor Pressure** | 17.05 mmHg @ 70°F | **Vapor Density** | 0.6 (Air = 1.0) |
| **Solubility** | Completely soluble | **Boiling Point/Range** | 212 °F / 100 °C |

## 10. STABILITY AND REACTIVITY

**Chemical Stability** Stable. Hazardous polymerization does not occur.

| | |
|---|---|
| **Conditions to Avoid** | None known |
| **Incompatible Products** | Strong oxidizing agents, Reducing agents, Amines, Powdered metals, Light and/or alkaline metals, Contact with metals liberates hydrogen gas. |
| **Hazardous Decomposition Products** | Carbon oxides, Nitrogen oxides (NOx), Sulfur oxides, Hydrogen chloride gas. |
| **Possibility of Hazardous Reactions** | None under normal processing |

## 11. TOXICOLOGICAL INFORMATION

**Product Information** No information available.

**Component Information**

**Acute Toxicity**

| Component | LD50 Oral | LD50 Dermal | LC50 Inhalation | Draize Test | Other |
|---|---|---|---|---|---|
| Magnesium nitrate | = 5440 mg/kg ( Rat ) | no data available | no data available | no data available | no data available |
| 5-Chloro-2-methyl-4-isothiazolin-3-one | = 481 mg/kg ( Rat ) | > 1008 mg/kg ( Rat ) | = 1.23 mg/L ( Rat ) 4 h | no data available | no data available |
| Magnesium chloride | = 2800 mg/kg ( Rat ) | no data available | no data available | no data available | no data available |

**Chronic Toxicity**

| Component | Mutagenicity | Sensitization | Developmental Toxicity | Reproductive Toxicity | Target Organ Effects |
|---|---|---|---|---|---|
| Magnesium nitrate | no data available | no data available | no data available | no data available | no data available |
| 5-Chloro-2-methyl-4-isothiazolin-3-one | no data available | Skin sensitization | no data available | no data available | Immune system |
| Magnesium chloride | no data available | no data available | no data available | no data available | no data available |

**Carcinogenicity**

| Component | ACGIH | IARC | NTP | OSHA | Other |
|---|---|---|---|---|---|
| Magnesium nitrate | not applicable | Group 2A | not applicable | X | not applicable |
| 5-Chloro-2-methyl-4-isothiazolin-3-one | not applicable | not applicable | not applicable | not applicable | not applicable |
| Magnesium chloride | not applicable | not applicable | not applicable | not applicable | not applicable |

## 12. ECOLOGICAL INFORMATION

**Product Information** No information available.

**Component Information**

| Component | Toxicity to Algae | Toxicity to Fish | Microtox | Water Flea | log Pow |
|---|---|---|---|---|---|
| Magnesium nitrate | no data available | no data available | no data available | no data available | N/A |
| 5-Chloro-2-methyl-4-isothiazolin-3-one | EC50 = 0.31 mg/L Anabaena flos-aquae 120 h<br>EC50 0.11 - 0.16 mg/L Pseudokirchneriella subcapitata 72 h<br>EC50 0.03 - 0.13 mg/L Pseudokirchneriella subcapitata 96 h | LC50 = 1.6 mg/L Oncorhynchus mykiss 96 h | EC50 = 5.7 mg/L 16 h | EC50= 4.71 mg/L 48 h<br>EC50 0.12 - 0.3 mg/L 48 h<br>EC50 0.71 - 0.99 mg/L 48 h | -0.71 - 0.75 |
| Magnesium chloride | EC50 = 2200 mg/L Desmodesmus subspicatus 72 h | LC50 = 4210 mg/L Gambusia affinis 96 h<br>LC50 1970 - 3880 mg/L Pimephales promelas 96 h | EC50 = 26140 mg/L 1 h<br>EC50 = 36300 mg/L 30 min<br>EC50 = 77200 mg/L 24 h | EC50= 1400 mg/L 24 h<br>EC50= 140 mg/L 48 h | N/A |

| | |
|---|---|
| **Persistence and Degradability** | No information available. |
| **Bioaccumulation** | No information available. |
| **Mobility** | No information available. |

## 13. DISPOSAL CONSIDERATIONS

| | |
|---|---|
| **Product Disposal** | Pesticide wastes are acutely hazardous. Improper disposal of excess pesticide or rinsate is a violation of federal law. If these wastes cannot be disposed of by use according to label instructions, contact your state pesticide or environmental control agency. |
| **Container Disposal** | Empty containers should be taken for local recycling, recovery, or waste disposal. Do not re -use empty containers. |

## 14. TRANSPORT INFORMATION

DOT

| | |
|---|---|
| **Proper Shipping Name** | Corrosive liquid, acidic, organic, n.o.s. |
| **Hazard Class** | 8 |
| **UN-No** | UN3265 |
| **Packing Group** | II |
| **Description** | Corrosive liquid, acidic, organic, n.o.s.(5 -chloro-2-methyl-4-isothiazolin-3-one),8,UN3265,PG II |

TDG

| | |
|---|---|
| **Proper shipping name** | Corrosive liquid, acidic, organic, n.o.s. |
| **Hazard Class** | 8 |
| **UN-No** | UN3265 |
| **Packing Group** | II |
| **Description** | Corrosive Liquid, Acidic, Organic, N.O.S.,.(5 -chloro-2-methyl-4-isothiazolin-3-one),8,UN3265,PG II |

ICAO

| | |
|---|---|
| **UN-No** | UN3265 |
| **Proper Shipping Name** | Corrosive liquid, acidic, organic, n.o.s.* |
| **Hazard Class** | 8 |
| **Packing Group** | II |
| **Shipping Description** | Corrosive liquid, acidic, organic, n.o.s.*(5 -chloro-2-methyl-4-isothiazolin-3-one),8,UN3265,PG II |

IATA

| | |
|---|---|
| **UN-No** | UN3265 |
| **Proper Shipping Name** | Corrosive liquid, acidic, organic, n.o.s.* |
| **Hazard Class** | 8 |
| **Packing Group** | II |
| **ERG Code** | 8L |
| **Shipping Description** | UN3265,Corrosive liquid, acidic, organic, n.o.s.*(5 -chloro-2-methyl-4-isothiazolin-3-one),8,PG II |

IMDG/IMO

| | |
|---|---|
| **Proper Shipping Name** | Corrosive liquid, acidic, organic, n.o.s. |
| **Hazard Class** | 8 |
| **UN-No** | UN3265 |
| **Packing Group** | II |
| **EmS No.** | F-A, S-B |
| **Shipping Description** | UN3265, Corrosive liquid, acidic, organic, n.o.s.(5 -chloro-2-methyl-4-isothiazolin-3-one,8,PG II |

## 15. REGULATORY INFORMATION

**Inventories**

| | |
|---|---|
| TSCA | Complies |
| DSL | Complies |

**U.S. Federal Regulations**

**SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

| Component | CAS-No | Weight % | SARA 313 - Threshold Values |
|---|---|---|---|
| Magnesium nitrate | 10377-60-3 | 1-5 | 1.0 |

**SARA 311/312 Hazardous Categorization**

| Acute Health Hazard | Chronic Health Hazard | Fire Hazard | Sudden Release of Pressure Hazard | Reactive Hazard |
|---|---|---|---|---|
| Yes | Yes | No | No | No |

**CERCLA**

| Component | Hazardous Substances RQs | CERCLA EHS RQs |
|---|---|---|
| Magnesium nitrate | Not applicable | Not applicable |
| 5-Chloro-2-methyl-4-isothiazolin-3-one | Not applicable | Not applicable |
| Magnesium chloride | Not applicable | Not applicable |

**Canada**

**This product may not be commercially placed on the market in Canada.**

**WHMIS Hazard Class**

Not applicable

## 16. OTHER INFORMATION

**Prepared By** Rachael Mohochi

| | |
|---|---|
| **Supercedes Date** | 12/13/2011 |
| **Issuing Date** | 07/09/2013 |
| **Reason for Revision** | No information available. |
| **Glossary** | No information available. |
| **List of References.** | No information available. |

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